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QUALITY MANAGEMENT SYSTEM PROCEDURE

Title: AMAZON ROBOTICS WELDING REQUIREMENTS	
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PROCESS OWNER: Hardware Engineering	Page 1 of 6
APPROVERS: TECHNICAL OPERATIONS, HARDWARE	

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1. Scope/Purpose

This document defines the minimum requirements for arc welding of steel and aluminum structures for Amazon Robotics.

Arc welds made on any Amazon Robotics assembly or weldment shall comply with the requirements of AWS D1.1 or D1.3 for steel structures, and AWS D1.2 for aluminum structures.

2. Applicability

- This document applies to the following arc welding processes:
 - Flux cored arc welding (FCAW)
 - Gas metal arc welding (GMAW), also known as MIG or MAG welding
 - Gas tungsten arc welding (GTAW), also known as TIG welding

3. Responsibility

- **Welder:** A person who performs manual or semiautomatic welding. Welders shall weld and inspect their welds.
- **Welding Operator:** A person who operates automated welding equipment. Welding operators may inspect their welds.
- **Welding Technician:** A person who sets up and performs routine adjustments to automated welding equipment. Welding technicians shall inspect their welds as part of the setup process for automated welding.
- **Welding Inspector:** A person who performs visual inspection of welds. The welding inspector shall inspect welds to meet the requirements of this document and applicable engineering standards and specifications.
- **Certified Welding Inspector (CWI):** A person certified by AWS as meeting the qualifications of AWS QC1 Standard for AWS Certification of Welding Inspectors. The CWI shall inspect welds, train welding inspectors, and supervise welding inspections.
- **Quality Engineer/Manager:** A person who establishes and monitors the Control Plan for the product.

4. Applicable Documents

- AWS D1.1 Structural Welding Code—Steel
- AWS D1.2 Structural Welding Code—Aluminum¹
- AWS D1.3 Structural Welding Code—Sheet Steel
- AWS D8.8M Specification for Automotive Weld Quality—Arc Welding of Steel
- AWS D8.14M Specification for Automotive Weld Quality—Arc Welding of Aluminum
- AWS B2.1 Specification for Welding Procedure and Performance Qualification
- AWS A2.4 Standard Symbols for Welding, Brazing, and Nondestructive Examination
- AWS A3.0 Standard Welding Terms and Definitions

¹ Section 4.12.3 of AWS D1.2 shall not apply to Amazon Robotics Weldments. Machining of plasma or laser cut edges is not required.

5. Welding Procedure Qualification

- Welding procedures shall be qualified to meet the requirements of the applicable code. For welding procedures qualified to meet the requirements of D1.3 for complete penetration groove welds, tensile tests shall also be required as listed in AWS B2.1.
- Each welding procedure specification (WPS) shall be supported by one or more procedure qualification records (PQR). Each PQR shall list all essential variables used to make the test weld and results of all required testing.
- Testing requirements for welding procedure qualification shall be defined by the joint design and further clarified by Amazon Robotics through the process control documents and/or drawings.
- A WPS shall be prepared to list acceptable ranges of all essential variables.
- Changes beyond the limitations of the essential variables shall require requalification of the WPS.
- AWS Standard Welding Procedure Specifications (SWPS) are acceptable for use.

6. Welding Performance Qualification

- Welders and welding operators shall be qualified for the applicable welding process to meet the requirements of the applicable referenced welding code for the material type being welded.
- A welder or welding operator performance qualification test record (PQTR) shall be created for each individual and application.
- Changes beyond the limitation of essential variables for welders and welding operators shall require requalification.

7. Weld Quality Controls

Weld quality controls monitor the conformance of weldments to the welding specifications. The engineering specification (weldment drawing) shall indicate the welding code that governs the weld. Weld quality control includes:

- Visual examination of welds
- Nondestructive dimensional evaluation of welds
- Examination of joint strength and/or fusion
- Examination records

a. Visual Examination of Welds

Welders and/or welding inspectors shall perform 100% visual examination of welds. Welding operators may also perform visual inspection of welds. Examples of discontinuities that may be identified through visual examination include craters, melt back, burn-through, porosity, undercut, nonmetallic inclusions, cracks, and notching.

b. Nondestructive Dimensional Evaluation of Welds

Nondestructive dimensional evaluation of welds shall be performed at first piece inspections, changeovers and setups, audit inspections, after all changes, and after all shift changes. Examples of issues that may be identified through nondestructive dimensional evaluation include missing, short, off location, and undersized welds.

c. Inspection for Adequate Joint Strength and/or Fusion

A weld joint may be examined for adequate joint strength or sufficient weld fusion. Requirements for examination of particular welds shall be specified on the product engineering specification (weldment drawing), and may include destructive (pry or chisel) testing (reference AWS D1.3), destructive metallographic (cut, polish, and etch) examination (Reference AWS D1.1 Figure 5.4, D1.2 Figure 4.2, D8.8M, D8.14M, and A3.0), or other methods that correlate to the engineering specification for the weldment.

d. Examination Records

Inspection records shall be maintained. The quality control of weld records do not need to be separate records; they may be maintained as part of a larger check sheet, process control sheet, control plan matrix or other standard procedure.

8. Process Capability and Control

See Material Approval Process Document 960-0015-05 for requirements detailing how vendors shall control their welding process and prove their process capable.

9. Welding Process Control Testing Requirements

See Material Approval Process Document 960-0015-05 for requirements detailing how vendors shall control their test their welds and weldments to prove their production product is meeting engineering requirements.

10. Standard Drawing Notes

e. Standard Drawing Note - Amazon Robotics Internal Reference only

See 960-0075 Engineering Drawing Standard for standard drawing notes for weldments

f. Visual Inspection Weld Notes - Amazon Robotics Internal Reference only

See 960-0075 Engineering Drawing Standard for standard visual inspection notes for weldments

11. Definitions

Definitions provided for document reference, see current code for most up to date definitions

- **Automatic Welding.** The equipment controls the electrode wire feeding and movement of the welding gun and/or workpiece.
- **Base Metal.** The material to be joined by welding.
- **Burn Through.** Excessive melt-through producing a hole.
- **Butt Joint.** A joint between two members aligned approximately in the same plane.
- **Contact Tip.** Transfers welding current from the power lead to the electrode.
- **Crack.** A fracture type discontinuity characterized by a sharp tip and high ratio of length and width to opening displacement.
- **Crater.** A depression at the termination of a weld.
- **Depth of Fusion.** The distance that fusion extends into the base metal or previous pass from the surface melted during welding. Depth of fusion is not the same as root or joint penetration.

- **Defect.** A discontinuity or discontinuities that by nature or accumulated effect render a part or product unable to meet minimum acceptable requirements or specifications. The term designates rejectability.
- **Discontinuity.** An interruption of the typical structure of a weld. A discontinuity is not necessarily a defect.
- **Effective Throat.** The minimum distance minus any convexity between the weld root and the face of a fillet weld.
- **Electrode.** (Weld Wire) The electrode transfers the current from the contact tube to the arc for FCAW and GMAW, and from the torch to the arc for GTAW.
- **Electrode Extension (Stickout).** While welding, the length of the unmelted electrode extending beyond the tip of the contact tip.
- **Essential Variables.** Conditions in which a change, as described in the specific variables, is considered to affect the mechanical properties of the weld joint.
- **Fillet Weld.** A weld of approximately triangular cross section joining two surfaces approximately at right angles to each other in a lap joint, T-joint, or corner joint.
- **Fillet Weld Leg.** The distance from the joint root to the toe of the fillet weld.
- **Fillet Weld Size.** For equal leg fillet welds, the leg lengths of the largest isosceles right triangle which can be inscribed within the fillet weld cross section. For unequal leg fillet welds, the leg lengths of the largest right triangle that can be inscribed within the fillet cross section.
- **Fusion.** The melting together of filler material and base metal or of base metal only which results in coalescence. Coalescence is the growing together or growth into one body of the materials being welded.
- **Incomplete Fusion.** (Cold Lap) A weld discontinuity in which fusion did not occur between weld metal and base metal or adjoining weld beads.
- **Joint Penetration.** The depth a weld extends from its face into a joint, exclusive of reinforcement.
- **Lap Joint.** A joint between two overlapping members in parallel planes.
- **Melt Back.** Melting of the base metal back from the edge that does not become part of the weld. This condition leaves a void between the weld deposit and the base metal.
- **Melt Through.** Visible root reinforcement produced in a joint welded from one side.
- **Notching (Edge Melt).** Gouging of the parent metal at the ends or edge of a welded joint.
- **Overlap (Cold Lap).** The protrusion of weld metal beyond the weld toe. Also a nonstandard term for incomplete fusion.
- **Penetration.** A nonstandard term for depth of fusion, joint penetration or root penetration. Penetration is not the same as depth of fusion.
- **Performance Qualification Test Record.** A record of welding variables used to produce an acceptable test weldment and the results of tests conducted on the weldment to qualify a welder or welding operator.
- **Porosity.** Cavity type discontinuities formed by gas entrapment during solidification.
- **Procedure Qualification Record (PQR).** A record of welding variables used to produce an acceptable test weldment and the results of tests conducted on the weldment to qualify a welding procedure specification.
- **Root Opening.** A separation at the joint root between the workpieces.
- **Root Penetration.** The depth that a weld extends into the joint root.

- **Semiautomatic Welding.** The equipment controls only the electrode wire feeding. The welding gun movement is controlled by hand.
- **Slag Inclusion.** Nonmetallic solid material entrapped in weld metal between weld metal and base metal.
- **Standard Welding Procedure Specification (SWPS).** A WPS published by AWS that is supported by PQRs reviewed and validated under the auspices of the Welding Research Council.
- **Theoretical Throat.** The distance from the beginning of the joint root perpendicular to the hypotenuse of the largest right triangle that can be inscribed within the cross section of a fillet weld. This dimension is based on the assumption that the root opening is equal to zero.
- **Throat.** See actual throat and effective throat.
- **Undercut.** A groove melted into the base metal adjacent to the weld toe or weld root and left unfilled by weld metal.
- **Weld Penetration.** A nonstandard term for root penetration or joint penetration.
- **Weld Toe.** The junction of the weld face and base metal.
- **Welding Procedure Specification (WPS).** A document providing the required welding variables for a specific application to assure repeatability by properly trained welders and welding operators.
- **Workpiece.** A part that is to be joined by welding.

REVISION HISTORY

Rev	ECO #	Description of Change	Sections Affected	Originator/Department
01	ECO-010640	Initial release	All	Michael O'Connor / Engineering
02	ECO-012693	Added footnote 1 to section 4	Section 4. Applicable Documents	Michael O'Connor / Engineering